TEAM 1302 · CSCI 3308

FlickPick

Swipe. Discover. Watch.

Course: CSCI 3308

Semester: Spring 2026

Team: 1302

THE TEAM

Meet Team 1302

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THE PRODUCT

What is FlickPick?

A swipe-based movie discovery app that makes finding your next favorite film fast, fun, and social.

Overwhelmed by endless streaming catalogs? FlickPick reimagines movie discovery — swipe right to save, left to pass, and let the app learn your taste over time.

Built for the 18–35 crowd who watches across Netflix, Hulu, HBO Max — and always struggles to pick something with friends.

Swipe to Discover

Browse movie cards with poster, rating, genre & synopsis. Right = save. Left = pass.

Personalized Recommendations

After 5+ swipes, FlickPick learns your genre, actor & director preferences.

Group Watch Sessions

Find movies the whole group has liked — end the "what should we watch?" debate.

Reviews & Social

Rate movies, write reviews, follow friends, and see what they're watching.

TECHNOLOGY

Tools & Technologies

	TOOL	CATEGORY	PURPOSE	RATING
	GitHub	VCS / Project Tracking	Version control, pull requests, issue tracking & branches	★★★★★
	PostgreSQL	Database	Relational DB — users, watchlists, swipe history, reviews	★★★★★
	Node.js + Express	App Server / Framework	REST API — routing, sessions, middleware & DB queries	★★★★★
	Handlebars	UI / Templating	Server-side HTML rendering for all pages and partials	★★★★☆
	TMDB API	External API	Movie metadata — posters, ratings, genres, cast & directors	★★★★★
	Mocha + Chai	Testing	Unit & integration tests for routes and business logic	★★★★☆
	Docker	Containerization	Reproducible dev environment with docker-compose	★★★★☆
	VS Code	IDE	Primary code editor for development across all team members	★★★★★
	Render	Deployment	Cloud deployment platform for the live production environment	★★★★☆

PROCESS

Development Methodologies

Iterative Development

Built features incrementally across weekly milestones, validating and refining each sprint before moving forward.

Agile Workflow

Weekly TA check-ins and sprint-based delivery kept the team aligned and adaptable to changing requirements.

Peer Code Reviews

All changes submitted via pull requests and reviewed by at least one teammate before merging into main.

Feature Branching

Each team member worked on dedicated branches — isolating features and preventing conflicts on the main codebase.

SYSTEM DESIGN

Architecture Diagram

PRESENTATION LAYER — CLIENT

Browser / Mobile Web
HTML · CSS · Vanilla JS

Handlebars Templates
Server-side rendering (.hbs)

Express Session
Cookie-based auth state

HTTP Requests / Rendered HTML

APPLICATION LAYER — NODE.JS + EXPRESS

Express Router
Routes & Controllers

Auth Middleware
bcryptjs · Session guard

Multer
Profile photo uploads

TMDB Cache
15-min in-memory TTL

SQL Queries via pg-promise

DATA LAYER — POSTGRESQL

users
Auth & profiles

watchlist
Saved & watched

swipe_history
Recommendations engine

reviews · friends
Social features

HTTPS REST Calls

EXTERNAL SERVICES

TMDB API
Movie data · Posters · Cast

Render
Cloud hosting & deployment

DOCKER

App + DB containerized via docker-compose

LESSONS LEARNED

Challenges & Solutions

01 TMDB API Rate Limits & Data Volume

Fetching fresh movie data for every user session risked hitting rate limits and caused slow load times when pulling from multiple categories.

Resolution: Implemented a 15-minute in-memory TTL cache per TMDB category/page combination

02 Building a Personalization Engine Without ML

Delivering meaningful recommendations required tallying genre, actor, and director preferences from swipe history — without a machine learning backend.

Resolution: Built a weighted frequency model — 30% genre-based, 30% person-based, 40% discovery

03 Docker + PostgreSQL Environment Consistency

Getting all five team members running the same dev environment with proper DB initialization was an early blocker that slowed onboarding.

Resolution: Standardized docker-compose setup with automated SQL init scripts for reproducible environments

WHAT'S NEXT

Future Enhancements

01

Streaming Platform Integration

Show users exactly where each movie is available to stream — Netflix, Hulu, HBO Max — directly on the movie card, eliminating the "where can I watch this?" friction entirely.

02

TV Show Toggle

Add a Movies / TV Shows switch so users can discover television series with the same swipe-based experience. Separate watchlists, ratings, and recommendations for each content type.

03

Native Mobile App

Port FlickPick to iOS and Android with true gesture-based swiping, push notifications for friend activity, and offline watchlist access for a premium native experience.

Live Demo

Let's see it in action

Click here for demo